Alberta Palaeontological Society

Speaker: Luke Nelson*

Time: September 18, 2026, 7:30 pm MDT

The anatomy, taxonomy, and tooth replacement of a sabre-toothed Cretaceous fish

Bio: Luke Nelson is a graduate of a master's and undergraduate degrees in evolutionary biology at the University of Alberta. He has worked on a number of projects with Dr. Alison Murray over the past five years, including his master's thesis work. Luke has a lifelong interest in fishes, which are the oldest and most diverse group of vertebrates. His research focuses on fishes of the Western Interior Seaway, which covered much of Alberta during the Cretaceous period.

Abstract:

Enchodus is an extinct genus of teleost fish that lived worldwide in the Late Cretaceous. The genus contains approximately 30 species, with six known from North America. Numerous *Enchodus* skulls were recovered in recent years from ammolite mines in the Bearpaw Formation of Southern Alberta, and most of these are referrable to *Enchodus petrosus* Cope, 1874. The most complete specimen is also the largest articulated skull yet described for the genus, resulting in an estimated 1.5 m total length for the animal. Another skull is preserved in three dimensions and was CT scanned, allowing a large fossa, which functions as a sheath for the first dentary fang on the premaxilla. With more than 20 specimens referable to *Enchodus* from the Tyrrell Museum collections, it is the most abundantly recovered group of fish from the Bearpaw Formation.

Enchodus dirus was named by Joseph Leidy based on a fragmentary dentary from North Dakota. Redescription of this holotype determines it to be nondiagnostic, lacking any characters to separate it from other species of *Enchodus*. A complete skull from Kansas, which was previously referred to *E. dirus*, is designated as a new species. This new species possesses a unique character suite, and sits close to other North American *Enchodus* species in a phylogenetic analysis. A nearly

complete skull from Greece, which was previously identified as *E.* cf. *E. dirus*, is not consistent with the characteristics of the new taxon.

Enchodus species have a single massive fang on the anteroventral corners of their paired palatine bones. I investigate the tissues and replacement of these fangs. These fangs are ankylosed to the bone, and are completely filled with a bone-like tissue known as osteodentine. The crescent-shaped bases of sequential teeth nest anteromedially to the previous tooth bases, and former bases are retained after the crown is shed. A wide review of CT scans of teeth from extant relatives of *Enchodus* is used to propose a potential tooth replacement mechanism in the genus. In this model, *Enchodus* replacement teeth initially grow in a horizontal orientation in a position medial to the palatine, then rotate 90 degrees to reach their functional position, prior to mineralization of their base. Several specimens appear to preserve replacement teeth undergoing this form of tooth replacement. Rotational tooth replacement is apparently advantageous for large-fanged fishes, since it prevents damage to growing replacement teeth before they emerge into the gape. Apparently for this reason, this form of tooth replacement has convergently evolved at least 3 times in ray-finned fishes.

* Please note that our speaker will be presenting remotely